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An Adaptive Cross-Resolution Network for Low-Resolution Person Re-Identification

Tariq Ali Arain [*,‡], Pengcheng Zhang [*,§], Sehrish Mazhar [†,¶] and Qing Meng [*,||]

[*]*Key Laboratory of Water Big Data Technology of Ministry of Water Resources*
College of Computer Science and Software Engineering, Hohai University
Nanjing 210098, Jiangsu, P. R. China

[†]*School of Mathematics, Hohai University*
Nanjing 210098, Jiangsu, P. R. China
[‡]*tariqkavisharain@yahoo.com*
[§]*pchzhang@hhu.edu.cn*
[¶]*sehrishwaraich6@gmail.com*
[||]*qmeng@hhu.edu.cn*

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Person re-identification (PReID) plays a critical role in intelligent surveillance systems, yet its performance is often hindered by the low resolution of real-world CCTV imagery. The mismatch between degraded probe images and high-quality gallery images introduces a cross-resolution gap that significantly reduces recognition accuracy. Addressing this challenge requires models capable of recovering discriminative visual cues while maintaining consistent feature representations across heterogeneous input resolutions. To this end, we propose VarReID, a multi-resolution-adaptable hybrid framework that integrates image super-resolution with a robust PReID architecture to overcome the limitations of low-resolution PReID (LR-PReID). The framework incorporates a lightweight generative adversarial network (GAN)-based super-resolution module with a De-noise Enhancement Block to reconstruct essential structural and textural details from degraded inputs. The super-resolved images are then processed by a ResNet-50-based PReID network enhanced with Random Erasing, Linear Warm-Up, and Circle Loss to support stable and discriminative identity learning. A Resolution Adjustment Unit (RAU) further standardizes input dimensions, ensuring consistent feature extraction across varying resolutions. This integration of SR and PReID enables VarReID to handle low-resolution inputs more effectively than previous LR-PReID methods. Its stability is demonstrated in ablation studies, maintaining 89.30% Rank-1 at 32×64, 82.63% at 16×32, and 52.43% at 8×16, and achieving 77.99% on the naturally low-resolution VR-Market dataset. Compared with recent strong LR-PReID models such as RAPSR+RAReID (73.70% Rank-1), VarReID attains superior retrieval accuracy (77.99% Rank-1 and 89.24% Rank-5) due to clearer structural restoration and more resolution-consistent feature representations. VarReID was evaluated on DukeMTMC-reID, VR-MSMT17, and VR-Market1501, achieving Rank-1 accuracies of 87.3%, 67.3%, and 77.99%, respectively, demonstrating its robustness under low-resolution conditions and its practical applicability.

[||]Corresponding author.

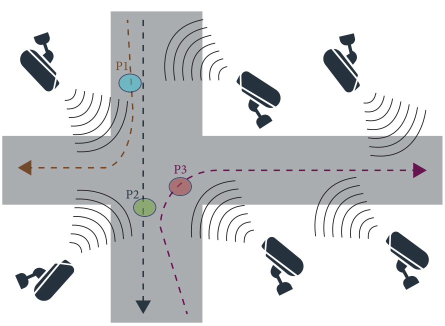
Keywords: Person re-identification (PReID); LR; resolution disparity issue; lower-resolution P-ReID (LR-P-ReID); and image enhancement.

1. Introduction

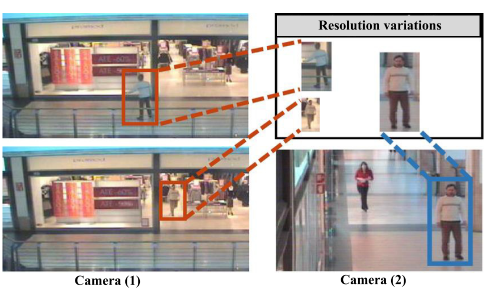
Person Re-Identification (PReID) is a critical task in computer vision and surveillance systems, enabling the tracking of individuals across nonoverlapping camera views over time. With the proliferation of CCTV networks in modern urban environments, vast amounts of video data are generated, making manual identity matching increasingly impractical. PReID addresses this challenge by automating person retrieval, thereby improving system reliability and operational efficiency.[2,51] As illustrated in Fig. 1(a), PReID supports continuous monitoring by matching individuals across disjoint camera feeds.

Despite its utility, PReID faces significant challenges, including variations in pose, viewpoint, illumination, and background clutter. Among these, resolution mismatch is particularly detrimental in real surveillance deployments. When a low-resolution (LR) probe image is matched against a high-resolution (HR) gallery image (or vice versa), as shown in Fig. 1(b), the resulting cross-resolution gap disrupts feature alignment and substantially reduces recognition accuracy.[38] In real-world scenarios such as shopping malls or public spaces, cameras with differing configurations often capture the same individual at varying resolutions. For example, a distant camera may produce LR images, whereas a nearby camera captures HR images of the same person. Such inconsistencies complicate robust feature matching and degrade overall PReID performance.

The low-resolution person re-identification (LR-PReID) problem is especially prevalent in surveillance systems, where many images are intrinsically LR due to hardware limitations, long-range viewpoints, or adverse environmental conditions. Consequently, maintaining consistent identification accuracy across heterogeneous



(a) PReID nonoverlapping cameras.



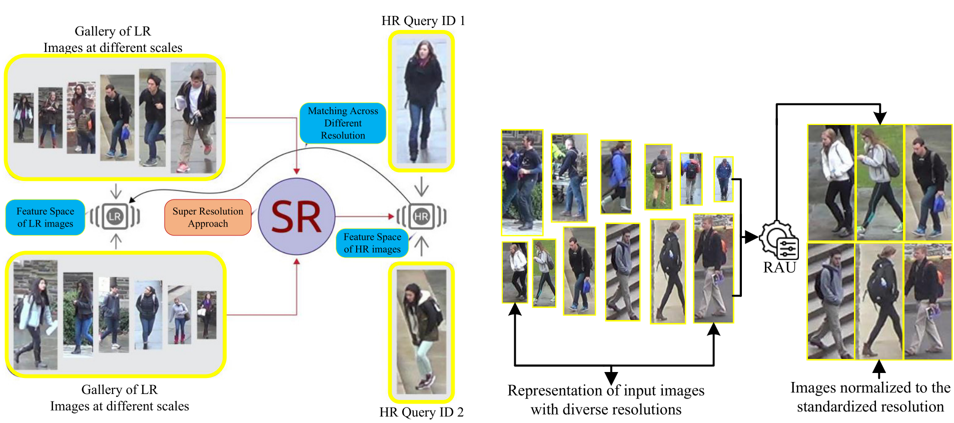
(b) Resolution gap in open-space PReID.[47]

Fig. 1. Illustration of challenges in cross-camera person re-identification. (a) Nonoverlapping camera setup. (b) Resolution variations in open-space PReID. The same individual is captured by two nonoverlapping cameras with different resolutions (LR versus HR), which complicates cross-camera matching due to reduced discriminative cues in LR images.

data sources becomes challenging, directly affecting the reliability of automated surveillance.[34] Addressing resolution variability is therefore essential for scalable and robust PReID in real deployments, where such variability is unavoidable.

Recent deep learning advances have significantly improved PReID on standard benchmarks.[41] However, many existing methods implicitly assume that probe and gallery images share similar resolution and visual quality. In practice, differences in pedestrian motion, camera distance, and compression lead to inconsistent resolutions and loss of fine-grained identity cues.[35] As shown in Fig. 2(a), resolution discrepancies between HR and LR inputs impair feature matching and complicate retrieval. This challenge is particularly evident in surveillance settings, where traditional methods have been reported to suffer up to a 19.2% drop in Rank-1 accuracy under cross-resolution evaluation.[19] Although multi-camera fusion has shown promise for related tasks such as anomaly detection,[24] existing PReID approaches remain limited in handling severe resolution mismatch.

Several approaches have been proposed to tackle LR-PReID.[14,41] However, these methods exhibit common limitations. First, many existing approaches project cross-resolution representations into a shared feature space; however, they often do not explicitly reconstruct discriminative details lost at LR, leading to weakened identity evidence under severe degradation.[48] Second, several methods relying on handcrafted descriptors or part-specific heuristics frequently exhibit limited robustness under shifts in dataset composition and camera characteristics compared with end-to-end deep models that learn resolution-invariant representations directly from data.[28] Third, many super-resolution (SR) pipelines adopt multi-branch designs with cascaded generative modules, such as CSR-GAN with three successive GAN units, incurring substantial memory and computational overhead and thus limiting deployment in resource-limited settings.[3] Recent work by Zhang *et al.*[45] shows that



(a) LR images at various resolutions. (b) Images after resolution standardization.

Fig. 2. Illustration of resolution variability and standardization in LR-PReID: (a) multi-scale low-resolution degradation; (b) resolution-standardized inputs.

task-driven SR can alleviate some of these issues; however, their focus on pose estimation differs from the identity-preserving requirements of PReID. Moreover, SR-based LR-PReID methods are often trained using synthetic downsampling and may fail to recover fine-grained texture when real surveillance data contain blur, compression artifacts, or sensor noise.[52] In such cases, SR models may hallucinate details that distort identity cues and destabilize downstream matching.[41] Meanwhile, Convolutional Neural Network-based (CNN) baselines typically struggle under ultra-low resolution because salient structural information is missing, and receptive-field aggregation becomes unreliable.[43] These limitations motivate hybrid designs that improve visual fidelity while enforcing resolution-consistent feature learning.

To address these challenges, we propose VarReID, a multi-resolution adaptable network for LR-PReID. VarReID integrates a lightweight GAN-based SR module with a D-noise (D-n) Enhancement Block and a Resolution Adjustment Unit (RAU) that standardizes heterogeneous input resolutions to a common scale, as illustrated in Fig. 2(b). The SR branch reconstructs structurally faithful images from ultra-low-resolution inputs, while the RAU encourages resolution-consistent feature alignment between probe and gallery images. This hybrid design yields more stable identity representations than prior LR-PReID frameworks, especially under severe degradation.

We focus on ultra-LR scenarios such as (8×16 and 16×32 pixels), where images are blurred and lack key identity cues such as edges, gradients, and color patterns. Using a ResNet-50 backbone, our PReID network further incorporates Random Erasing (RE) and Linear Warm-Up (LW-up) to improve generalization under low-quality inputs and Circle Loss (CL) to enhance discriminative learning. VarReID achieves Rank-1 accuracies of 87.3% on MLR-DukeMTMC-ReID, 67.3% on VR-MSMT17, and 77.99% on VR-Market1501. In addition, ablation studies confirm consistent gains across multiple resolution levels, demonstrating robustness under extreme resolution reduction. Our key contributions are summarized as follows:

- We introduce VarReID, a multi-resolution-adaptable hybrid framework that improves PReID under ultra-low-resolution and cross-resolution conditions.
- We propose an integrated hybrid architecture that combines a lightweight GAN-based SR module, a De-noise Enhancement Block, and RAU to restore structural details and standardize heterogeneous input resolutions.
- We perform extensive experiments and ablation studies across multiple benchmarks, including MLR-DukeMTMC-reID, VR-MSMT17, and VR-Market1501, demonstrating VarReID's effectiveness and stability, achieving up to 89.30% Rank-1 accuracy under 32x64 low-resolution settings.

2. Related Work

This section reviews recent studies on LR-PReID and image enhancement techniques relevant to identity-preserving reconstruction.

2.1. *Low-Resolution (LR) Person Re-Identification (PReID)*

PReID aims to recognize persons across nonoverlapping camera views despite challenges such as pose variations and occlusions.[43] Significant progress has been made in addressing these challenges, such as the work by Yang *et al.*,[40] Yuan *et al.*,[42] and Du *et al.*[7]; however, a major practical challenge remains the low resolution of many real-world surveillance images. As a result, the LR-PReID issue has gained increasing attention.

Several techniques have been proposed to reduce the disparity between LR and HR images.[35] However, many early approaches assume access to HR gallery images, which is often unrealistic in operational surveillance settings. Ghadekar *et al.*[9] utilized conditional GANs to develop a dual-generator and dual-discriminator architecture that improved image quality and identity preservation, but at the cost of increased computational complexity. Adil *et al.* [1] combined SR and PReID models by integrating ESRGAN to enhance LR inputs and improve identification accuracy. Santana *et al.*[26] showed that deep learning–based image enhancement can significantly improve PReID performance under degraded visual conditions, supporting the use of restoration modules, for example, GAN-based SR and denoising for LR-PReID.

Recent GAN-based LR-PReID research has further advanced resolution enhancement and identity preservation through domain-aware and resolution-adaptive super-resolution strategies. Multi-domain SR-GANs have been proposed to model degradation-specific mappings more effectively,[43] while perception–distortion balanced SR frameworks improve both fidelity and realism during reconstruction.[30] Semi-supervised and unsupervised GAN-based SR models have also emerged, enabling the use of large-scale unlabeled surveillance data to enhance low-quality person images.[21] Despite these advances, performance remains limited at extremely low resolutions, including, 8×16, where identity-critical texture, contour, and color cues are severely degraded. Motivated by this gap, our work adopts a lightweight hybrid SR design tailored to LR-PReID to better recover structural cues while preserving identity information.

2.2. *High-Resolution (HR) Image Reconstruction Techniques*

The image SR is an essential technique in computer vision that seeks to reconstruct HR images from LR counterparts. It has attracted significant attention due to its application in several sectors, such as surveillance, medical imaging, and remote sensing.[15] The Single-Image Super-Resolution (SISR) has significantly developed, notably with the use of CNNs.

GAN-based super-resolution further improves perceptual fidelity by encouraging realistic high-frequency detail reconstruction. Classical models such as SRGAN established the foundation for adversarially driven high-frequency reconstruction,[17] while more recent advances have expanded this direction. For example, evolutionary multi-objective optimization has been used to better balance perception–distortion trade-offs,[30] and distillation-based frameworks have shown strong adaptation to real-world degradations without paired data.[44] Furthermore, enhanced GAN architectures — including ConvNeXt-based discriminators[5] and semantic-prior–guided generators[33] — demonstrate notable gains in texture realism and structural consistency. Collectively, these studies highlight the value of adversarial and perceptual learning for restoring discriminative details that benefit downstream recognition tasks such as PReID.

More recently, Park *et al.* [23] proposed an SISR framework that converts LR inputs into high-quality HR outputs. Inspired by Park *et al.*, we developed a GAN-based generative approach to strengthen our HR reconstruction component. By learning an identity-preserving restoration prior, the enhancement module produces clearer structural cues that facilitate more reliable feature extraction in the subsequent PReID network.

3. Proposed Model Architecture

This section describes the VarReID architecture and its constituent modules for LR and cross-resolution PReID. The VarReID model consists of three primary components: (1) a RAU, (2) an SR network, and (3) a PReID network.

(1) The RAU initially processes input images with varying resolutions. This unit normalizes the image dimensions to a standardized size, as outlined in Table 1, which enables subsequent processing. Let r denote the downsampling factor; the adjusted image dimensions are (W/r) and (H/r). To accommodate different input resolutions during training, the RAU uses bicubic interpolation to resample images to a dataset-specific normalized resolution before passing them to the SR Network. Concretely, given an input RGB image $\mathbf{I} \in \mathbb{R}^{H\times W\times 3}$, the RAU determines a target resolution (H_n, W_n) such as 8×32 for VR-Market1501, 16×32 for VR-MSMT17, or 30×60 for MLR-DukeMTMC-reID as specified in Table 1 and resamples $\mathbf{I}$ to this canonical size using a bicubic interpolation operator $\mathcal{B}$. The normalized image is

$$\tilde{\mathbf{I}} = \mathcal{B}(\mathbf{I}) \in \mathbb{R}^{H_n\times W_n\times 3}, \tag{1}$$

where for a single channel, each pixel value $\tilde{I}(u,v)$ is obtained by

$$\tilde{I}(u,v) = \sum_{m=-1}^{2}\sum_{n=-1}^{2} I(x_0+m,\, y_0+n)\, k(\alpha-m)\, k(\beta-n), \tag{2}$$

with

$$x = \frac{W}{W_n}v, \quad y = \frac{H}{H_n}u, \quad x_0 = \lfloor x \rfloor, \quad y_0 = \lfloor y \rfloor, \quad \alpha = x - x_0, \quad \beta = y - y_0 \tag{3}$$

and $k(\cdot)$ denoting the cubic convolution kernel. This normalization ensures that all images within a mini-batch share the same spatial dimensions and delivers resolution-consistent inputs to the subsequent SR Network.

(2) The second module converts LR images into HR images via the SR network. The SR-Network comprises two components: (1) the main element is the generative model (G_ϕ), and the second component is the discriminator. The main functionality of the generative model (G_ϕ) is to take input (x) as an LR image and generate (y) as an HR image according to Eq. 5. The Discriminator distinguishes between real HR images and those generated by (G_ϕ). Moreover, a D-n module is employed to enhance image quality before ReID feature extraction.

In our implementation, the D-n module operates *after* the SR generator G_ϕ produces the super-resolved image $\mathbf{I}_{\mathrm{SR}}$. Specifically, we apply a color nonlocal means denoising filter that removes amplified noise and compression artifacts introduced during SR. For each pixel location (u, v), the cleaned output $\mathbf{I}_{\mathrm{clean}}$ is obtained as

$$\mathbf{I}_{\mathrm{clean}}(u, v) = \sum_{(i,j)\in\Omega_{u,v}} w_{(u,v),(i,j)}\, \mathbf{I}_{\mathrm{SR}}(i, j), \tag{4}$$

where $\Omega_{u,v}$ denotes a search window around (u, v), and the weights $w_{(u,v),(i,j)}$ are normalized such that $\sum w_{(u,v),(i,j)} = 1$ and decrease with the squared patch distance between color neighborhoods centered at (u, v) and (i, j). This denoising stage suppresses noise, spurious high-frequency textures, and SR-induced artifacts while preserving critical identity cues such as edges, contours, and clothing structures. The resulting image $\mathbf{I}_{\mathrm{clean}}$ is then forwarded to the PReID backbone for robust feature extraction.

(3) The VarReID network's final stage involves feeding the HR images generated by the generative model (G_ϕ) of the SR network part to the PReID head. This PReID network is based on the ResNet-50 architecture, which extracts discriminative features and allows for accurate PReID. The overall pipeline is illustrated in Fig. 3. Algorithm 1 provides the pseudo-code for the proposed VarReID network, which includes a RAU and feature extraction. The following subsections describe the SR Network and the PReID network in detail.

3.1. *Image Super-Resolution (SR)*

The SR network adopts a structured framework to enhance LR images into high-quality super-resolved outputs. As illustrated in Fig. 3, the workflow depicts how the components interact, together with their associated loss functions. The architecture comprises two main elements: a generative model and a discriminator. The generative model $G_\theta(x)$ takes the LR input x and produces a HR estimate $\hat{y}$. This model exploits intrinsic LR features to preserve fine structures and improve overall visual quality. The input–output relationship is defined in Eq. 5.

$$\hat{y} = G_\theta(x_{\mathrm{LR}}), \tag{5}$$

where $\hat{y}$ denotes the SR image, x_{LR} represents the LR input, and θ represents the learnable parameters of the generative model G.

The SR network optimizes the generated SR images using a carefully designed total loss function to ensure that they closely match the HR ground truth. The

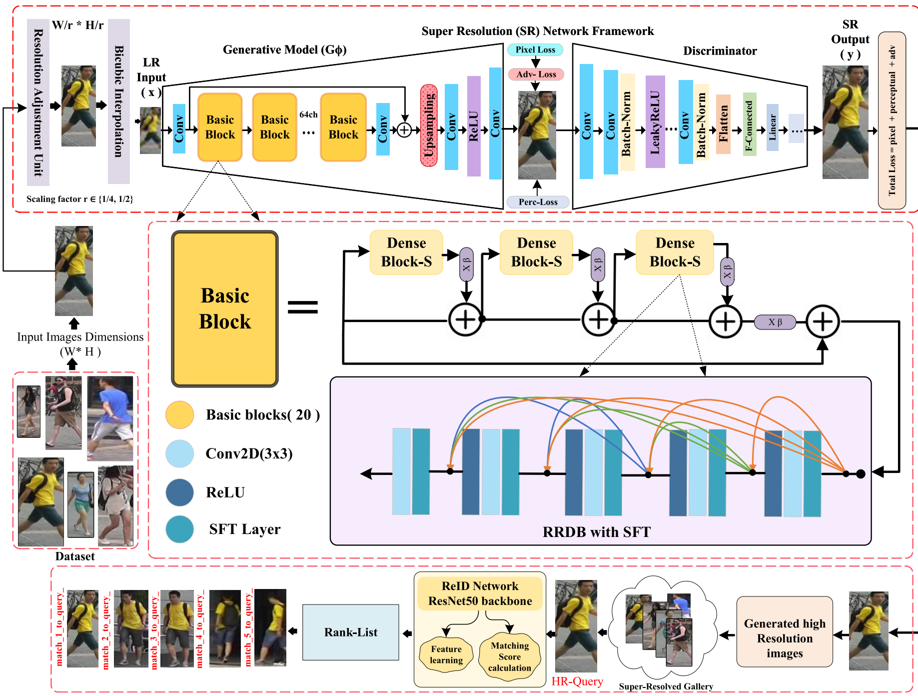


Fig. 3. The complete illustration of the proposed PReID network.

architecture includes a sequence of residual-in-residual dense blocks (RRDBs), each incorporating spatial feature transform (SFT) layers, convolutional units, and activation functions. This configuration enables the model to extract rich spatial features from LR inputs while learning the residual differences between LR and HR images. By focusing on these residual discrepancies, the network produces high-quality SR outputs.

The core of the generative model G_θ is the RRDB module, which consists of 23 basic blocks. These blocks are designed to extract meaningful features and preserve important visual details during reconstruction. Through these components, the SR network generates super-resolved images that retain structural fidelity and identity-relevant information.

The three loss functions, reconstruction loss ($\boldsymbol{L}_{\text{Rec}}$), adversarial loss ($\boldsymbol{L}_{\text{adv}}$), and perceptual loss ($\boldsymbol{L}_{\text{per}}$), are minimized and combined to compute the total loss:

$$L_{\text{Total}} = L_{\text{Rec}} + L_{\text{Per}} + L_{\text{Adv}}. \tag{6}$$

Here, L_{Rec} enforces pixel-level similarity between the SR and HR images, ensuring that the generated output preserves the underlying structure and fine details. L_{Per} preserves high-level visual characteristics by comparing deep feature representations extracted from a pre-trained network such as VGG, encouraging the generator to retain meaningful textures, edges, and contours. L_{Adv}, derived from adversarial learning, drives the generator to produce photo-realistic images that can fool the discriminator, reducing blurriness and overly smooth artifacts. Combined, these loss terms guide the SR network to generate images that are both structurally accurate and visually natural. Furthermore, the generative model, loss functions, the discriminator, and adversarial training are discussed in the following sections.

- **Generative (G_θ) Model and Loss Functions:** The generative model $G_\theta(x|\psi)$ takes an LR image x as input and predefined prior information ψ, generating a super-resolved image $\hat{y}$. The model improves output quality by extracting features from the LR image and the prior information. The three training components minimize the overall loss $L(\hat{y}, y, \lambda)$: L_{Rec} (Eq. (8)), L_{per1} (Eq. (9)), and L_{adv} (Eq. (10)).
$$L(\hat{y}, y, \lambda) = \lambda_1 L_{\text{Rec}} + \lambda_2 L_{\text{per1}} + \lambda_3 L_{\text{adv}}, \tag{7}$$
where $\lambda_1, \lambda_2, \lambda_3$ are the weights for each loss component, thus guiding the optimization procedure of the generative model G_θ. The individual loss components are defined as follows:
$$L_{\text{Rec}} = \frac{1}{N} \sum_{i=1}^{N} \|\hat{y}_i - y_i\|_2^2, \tag{8}$$
where N represents the total number of pixels, $\hat{y}_i$ is the predicted pixel value at position i, and y_i corresponds to the ground truth pixel value at the same position.
$$L_{\text{per1}} = \frac{1}{M} \sum_{j=1}^{M} \|\phi_j(\hat{y}) - \phi_j(y)\|_2^2, \tag{9}$$
where M denotes the number of feature maps, and $\phi_j(\cdot)$ represents the feature extraction method at layer j of the pre-trained VGG network.
$$L_{\text{adv}} = -\mathbb{E}_y[\log(D(y))] - \mathbb{E}_{\hat{y}}[\log(1 - D(\hat{y}))], \tag{10}$$
where $D(\cdot)$ is the discriminator function that distinguishes between authentic images y and generated images $\hat{y}$.
$$\theta^* = \arg\min_\theta E_{x,y \sim p_{x,y}}[L(\hat{y}, y, \lambda)]. \tag{11}$$
where θ represents the learnable parameters of the generative model G_θ, and the expectation is computed over the training dataset consisting of pairs (x, y).

- **Discriminator and Adversarial Training:** The discriminator VGG-128 is used to minimize the adversarial loss L_{adv} by differentiating between HR images and SR images produced by G_θ. The adversarial loss L_{adv} is defined as follows:
$$L_{\text{adv}} = -\mathbb{E}_{y_{\text{HR}}}[\log(D(y_{\text{HR}}))] - \mathbb{E}_{\hat{y}}[\log(1 - D(\hat{y}))]. \tag{12}$$
where $D(\cdot)$ is the discriminator function, y_{HR} is the original HR image, and $\hat{y}$ is the generated super-resolved image.

The generator is encouraged to produce increasingly difficult images for the discriminator to distinguish from real HR images by minimizing this loss. To further enhance perceptual quality, we incorporate a VGG-based feature extractor to compute ($\boldsymbol{L}_{\text{per}}$):

$$L_{\text{per1}} = \sum_{j=1}^{M} \|\phi_j(y_{\text{HR}}) - \phi_j(\hat{y})\|_2^2, \tag{13}$$

Algorithm 1. VarReID Network

Require: LR images $\mathbf{I}_{LR}$, Feature representations from gallery images $\mathbf{F}_{gallery}$
Ensure: Rank-ordered list of matched identities $\mathbf{R}$

Step 1: Resolution Adjustment
for each image in $\mathbf{I}_{LR}$ **do**
    $\mathbf{I}_{adj} \Leftarrow$ ResolutionAdjustment(image, r) ▷ Scale down using $r \in \{1/4, 1/2\}$
end for

Step 2: Bicubic Interpolation
$\mathbf{I}_{bicubic} \Leftarrow$ BicubicInterpolation($\mathbf{I}_{adj}$) ▷ Interpolate the adjusted images

Step 3: SR Network
3.1 Generative Model (G):
Generate the super-resolved image $\hat{y}$ from the LR input x_{LR}:

$$\hat{y} = G_\theta(x_{\text{LR}}) \tag{14}$$

3.2 Discriminator:
Distinguish between ground truth HR images and those generated by G to produce the super-resolved image $\hat{y}$, using the Discriminator VGG_{128} architecture.

Step 4: D-noise
$\mathbf{I}_{D-noise} \Leftarrow$ D-noise($\mathbf{I}_{SR}$)

Step 5: PReID-Network
5.1 Feature Extraction from HR Image:
$\mathbf{F}_{reid} \Leftarrow$ FeatureExtractor($\mathbf{I}_{denoise}$)
5.2 Matching Score Calculation:
$\mathbf{S}_{match} \Leftarrow$ MatchScore($\mathbf{F}_{reid}$, $\mathbf{F}_{gallery}$)

Step 6: Rank List Generation
$\mathbf{R} \Leftarrow$ GenerateRankList($\mathbf{S}_{match}$)

Return: Rank-ordered list $\mathbf{R}$

where ϕ_j represents feature extraction at layer j of a pre-trained VGG network, and M is the number of feature maps used for comparison. This (L_{per}) ensures that the generated SR images resemble authentic HR images.

3.2. *Person Re-Identification*

For PReID, we adopt a ResNet-50 backbone, initialized with Kaiming initialization and fine-tuned from ImageNet pretrained weights to leverage transferable visual representations.[11] To improve generalization and robustness under low-quality inputs, we incorporate three training strategies: RE, LW-up, and CL. These components are used throughout training to stabilize optimization and enhance discriminative feature learning.

- **Random Erasing (RE):** Random Erasing, proposed by Zhong *et al.*,[50] improves robustness by randomly selecting a rectangular region and replacing its pixels with random values (or the dataset mean) with probability p (see Fig. 4). This augmentation is analogous to dropout in the input space: it prevents the network from over-relying on a few localized cues and encourages learning more complete and diverse identity evidence. In our implementation, we apply an image- and object-aware variant of RE, which erases regions on the foreground person as well as background areas, improving resilience to occlusion and partial observations.
- **Linear Warm-Up (LW-up):** To ensure stable training, we apply LW-up, which increases the learning rate from an initial value α_{initial} to a target value α_{target} over T warm-up epochs. The learning rate at epoch t is defined as:

$$\alpha(t) = \alpha_{\text{initial}} + \left(\frac{\alpha_{\text{target}} - \alpha_{\text{initial}}}{T}\right)t. \tag{15}$$

Fig. 4. RE replaces randomly selected regions with random pixel values.

Following the linear-scaling principle, the target learning rate is chosen according to the batch size scaling factor k. Warm-up avoids abrupt learning-rate changes early in training, improves convergence stability, and benefits feature learning under resolution-degraded inputs.

- **Circle Loss (CL):** We employ CL to enhance embedding discriminability by promoting intra-class compactness and inter-class separation, while placing higher emphasis on hard positive and hard negative pairs. The loss is formulated as

$$\mathcal{L} = \frac{1}{N}\sum_{i=1}^{N}\sum_{j=1}^{N}(y_{ij}\max(0, m_1 + \Delta_i - \cos\theta_{ij}) + (1 - y_{ij})\ \max(0, \cos\theta_{ij} - m_2)), \tag{16}$$

where $\cos(\theta_{ij})$ denotes the cosine similarity between embeddings $\mathbf{x}_i$ and $\mathbf{x}_j$, and $y_{ij} \in \{0, 1\}$ indicates whether samples i and j share the same identity. Here, m_1 and m_2 are margins for positive and negative pairs, respectively, and Δ_i is an anchor-dependent adaptive margin. By weighting harder pairs more strongly, Circle Loss improves identity separation and consistently boosts PReID retrieval performance.

4. Experimental Setup

This section describes the experimental setup and evaluation protocol for VarReID. We report the datasets used (VR-Market1501, VR-MSMT17, and MLR-DukeMTMC-reID) and the corresponding training and implementation settings. Table 1 provides an overview of dataset statistics.

4.1. *Datasets*

VR-Market1501 is a multi-resolution variant of Market1501, collected in front of a supermarket at the Tsinghua University. The standard split is preserved (751 training and 710 testing identities). Images are downsampled to widths from 8 to 32 pixels, resulting in 24 resolution settings.

VR-MSMT17 is derived from MSMT17, a large-scale PReID benchmark captured by a 15-camera network under diverse conditions. We use its various-resolution version by downsampling images to multiple widths, yielding 96 resolution settings, while keeping the original identity split and image distribution unchanged.

Table 1. Dataset statistics including the number of identities, train/test splits, total images, and LR input size.

Dataset	IDs	Train	Test	Total images	LR-size
VR-Market1501	1501	7518	7518	32 217	$[8, 32]$
VR-MSMT17	4101	1042	2,246+807	126 411	$[16, 32]$
MLR-DukeMTMC-reID	1812	702	702+408	36 441	$[30, 60]$

MLR-DukeMTMC-reID follows the Market1501 evaluation protocol and is constructed from DukeMTMC. It includes 16 522 training images from 702 identities, 2228 query images from the same 702 identities, and 17 661 gallery images containing 702 identities plus 408 distractors. Low-resolution inputs are generated according to the ranges in Table 1.

Compared with standard PReID benchmarks, these VR settings are more challenging because probe and gallery images have heterogeneous resolutions, which increases intra-class variation and exacerbates cross-resolution mismatch.

4.2. *VarReID training*

The training procedure is organized into three stages. First, all input images are resized to a dataset-specific canonical resolution using the RAU, ensuring consistent spatial dimensions for subsequent modules. This normalization stabilizes optimization and makes the feature extraction process resolution-consistent across the training set.

Second, the normalized images are downsampled using MATLAB bicubic interpolation to simulate resolution variations and generate LR inputs. These LR images are used to train the SR network with a mini-batch size of 8. We optimize the SR model using Adam with $(\beta_1 = 0.9, \beta_2 = 0.99)$ and a learning rate of 1×10^{-4}, together with a MultiStepLR schedule that decays the learning rate by a factor of 0.5 at 12 500, 25 000, and 37 500 iterations. The SR objective combines RaGAN, L1, and perceptual loss ($\boldsymbol{L}_{\text{per}}$), and checkpoints are saved during training. Finally, the PReID network is fine-tuned using HR images produced by the trained SR generator G_θ. We adopt a ResNet-50 backbone initialized from ImageNet weights and train the PReID model in PyTorch on a single NVIDIA Tesla V100 GPU with batch size 8 for 100 epochs. The learning rate is set to 0.02 with a linear warm-up over the first 5 epochs. To improve robustness under ultra-low-resolution conditions, we apply RE with probability 0.5 and dropout with rate 0.5. We use CL to strengthen discriminative embedding learning.

For reproducibility, the key PReID training settings are: batch size 8, learning rate 0.02, warm-up epochs 5, total epochs 100, weight decay 5×10^{-4}, stride 1, and random erasing probability 0.5. All experiments are conducted with single-GPU training (GPU id 0).

5. VarReID Evaluation

This section reports two sets of experiments: (1) qualitative evaluation of SR image generation and (2) quantitative evaluation of PReID performance on four benchmarks. We use two standard metrics: mean Average Precision (mAP), which measures overall retrieval performance, and the Cumulative Matching Characteristic (CMC) at top-K, which evaluates rank-based matching accuracy. Section 5.2 provides a detailed comparison with state-of-the-art (SOTA) methods.

5.1. *SR network evaluation*

This subsection evaluates the SR network's capability to reconstruct HR images from LR inputs. We visually compare four outputs: the LR input, the SR result, the refined SR+D-n result, and the HR ground truth, as shown in Fig. 5. Although the SR generator restores fine structures, it may also introduce amplified noise or compression-related artifacts that reduce visual clarity and can degrade downstream PReID feature extraction. Applying the proposed D-n module suppresses these artifacts, yielding sharper details and more reliable identity cues.

5.2. *Comparison of SOTA to VarReID models*

This section presents a comprehensive comparison between the proposed VarReID framework and representative SOTA methods for LR-PReID. We evaluate all methods under a unified protocol on three widely adopted benchmarks, MLR-DukeMTMC-reID, VR-Market1501, and VR-MSMT17, using CMC Rank-1 and Rank-5 accuracy as the primary evaluation metrics. To ensure a fair comparison, we focus the SOTA analysis on CMC metrics, since mAP values are not consistently reported across prior works under identical experimental settings. For completeness, VarReID's mAP scores under our protocol are reported separately in Table 5.

MLR-DukeMTMC-reID. On the MLR-DukeMTMC-reID benchmark, VarReID achieves 87.30% Rank-1 and 94.20% Rank-5 accuracy, as shown in Table 2, demonstrating strong retrieval performance under cross-resolution conditions.

Fig. 5. Qualitative SR results for VarReID. From left to right: LR input, SR output, refined SR+D-n output, and HR ground truth. Yellow borders indicate model outputs and red borders indicate HR references.

Table 2. The results of MLR-DukeMTMC.

Method	Rank-1%	Rank-5%
SING[15]	65.20%	80.10%
PAN[49]	71.60%	—
PN-GAN[25]	73.60%	—
SVDNet[31]	76.70%	—
HA-CNN[18]	80.50%	—
MLFN[4]	81.00%	—
KPM[27]	80.30%	—
PABR[29]	84.40%	—
PCB-RPP[32]	83.30%	—
CSR-GAN[35]	77.20%	88.10%
MSA-SR-PREID[1]	79.06%	90.00%
CAD-Net++[20]	77.20%	88.10%
PyrNet+PRI[10]	82.10%	91.10%
SAN[16]	87.00%	—
LRAN[38]	89.60%	—
BPBreID[28]	89.60%	—
CRReID[37]	81.90%	92.40%
VarReID	**87.30%**	**94.20%**

Compared with SR-based and resolution-aware baselines such as CSR-GAN,[35] MSA-SR-PREID,[1] and FD-GAN,[8] VarReID consistently delivers higher Rank-5 accuracy, indicating improved robustness when the correct identity must be retrieved from a small candidate set rather than a single top match. Although LRAN[38] and BPBreID [28] report higher Rank-1 accuracy, VarReID exhibits stronger top-K stability, suggesting that the proposed SR and resolution-adjustment pipeline enhances feature consistency under severe resolution degradation. This behavior aligns with real-world surveillance scenarios, where exact Rank-1 matching is often ambiguous due to extreme low resolution and viewpoint variation.

VR-Market1501. On the VR-Market1501 dataset, VarReID attains 77.99% Rank-1 and 89.24% Rank-5 accuracy, as reported in Table 3. These results outperform several recent LR-PReID approaches, including LtH-MRCN,[7] DRSL,[39] RV loss,[42] and RAPSR+RAReID.[41] While methods such as UniCat[6] and CAD-Net++ [20] achieve higher Rank-1 accuracy, VarReID remains competitive and demonstrates consistently strong Rank-5 performance across heterogeneous resolutions. This indicates that VarReID effectively balances discriminative power and robustness, particularly when gallery images exhibit diverse resolution levels.

VR-MSMT17. VR-MSMT17 represents a more challenging large-scale scenario with substantial resolution diversity and environmental variation. Under this setting, VarReID achieves 67.30% Rank-1 and 78.70% Rank-5 accuracy, as presented in Table 4, outperforming classical resolution-aware baselines such as RIPR (FFSR +RIFE) [22] and recent SR-based methods including RAPSR+RAReID.[41] The consistent gains on VR-MSMT17 suggest improved generalization of VarReID under real-world multi-resolution surveillance conditions, where both identity appearance and image quality vary significantly across cameras.

Table 3. The quantitative results for the VR-Market1501 dataset.

Method	Rank-1%	Rank-5%
DenseNet121[13]	60.00%	78.80%
SE-ResNet50[12]	58.20%	78.60%
SING[15]	60.50%	81.80%
CSR-GAN[35]	59.80%	81.30%
FFSR[22]	59.20%	80.10%
MSA-SR-PREID[1]	68.26%	85.71%
UniCat[6]	83.80%	93.50%
CAD-Net++[20]	84.10%	93.00%
PFNet[46]	83.60%	92.80%
MMIM[36]	84.60%	94.20%
LtH-MRCN[7]	71.40%	86.90%
DRSL[39]	73.80%	88.30%
RV loss[42]	72.60%	89.50%
MFDNet[34]	74.20%	—
RAPSR+RAReID[41]	73.70%	88.50%
VarReID	**77.99%**	**89.24%**

Table 4. The quantitative results for the VR-MSMT17 dataset.

Method	Rank-1%	Rank-5%
DenseNet121[13]	51.20%	67.40%
SE-ResNet50[12]	52.30%	68.90%
SING[15]	52.10%	68.30%
CSR-GAN[35]	51.90%	67.50%
FFSR[22]	52.80%	69.00%
RIFE[22]	53.30%	70.10%
RIPR (FFSR+RIFE)[22]	55.50%	72.40%
MSA-SR-PREID[1]	60.65%	75.20%
RAPSR+RAReID[41]	61.50%	77.40%
VarReID	**67.30%**	**78.70%**

Across all three benchmarks, VarReID demonstrates consistently strong Rank-5 accuracy together with competitive Rank-1 performance, highlighting the effectiveness of jointly integrating super-resolution, resolution adjustment, and identity-aware feature learning. Rather than optimizing exclusively for top-1 retrieval, VarReID improves the quality and stability of the candidate set under severe resolution mismatch, which is critical for practical PReID systems. The reported mAP scores in Table 5 further confirm VarReID's strong retrieval precision under the proposed evaluation protocol.

5.3. *VarReID: Performance and challenges*

As illustrated in Fig. 10, VarReID demonstrated SOTA performance, consistently outperforming existing methods across multiple benchmarks. The model exhibited

Table 5. mAP scores and Rank-1 Accuracy for the proposed VarReID model.

Dataset	mAP Score	Rank-1 (%)
MLR-DukeMTMC-reID	83.60%	87.30%
VR-Market1501	67.30%	77.99%
VR-MSMT17	41.20%	67.70%

particularly strong results at moderate scales, achieving a Rank-1 accuracy of 89.3% at 32×64. However, its performance degraded substantially under ultra-low-resolution conditions, with Rank-1 dropping to 52.4% at 8×16. This sharp decline underscored the inherent difficulty of recovering discriminative identity cues from severely compressed imagery. These results support the effectiveness of integrating SR and denoising for handling realistic resolution variability, while also exposing clear limitations under extreme compression. Despite these challenges, VarReID maintained robust performance under variable-resolution settings, attaining a Rank-1 accuracy of 77.99%, thereby surpassing competing methods in conditions that closely resemble real-world surveillance. Nonetheless, the pronounced degradation at extreme resolutions highlighted important directions for future work, particularly in improving feature recovery and representation learning under severe information loss.

6. Ablation Work

This section presents a comprehensive ablation study to demonstrate the novelty and significance of our proposed VarReID model for PReID. We evaluated the benefits of integrating SR techniques and optimizing learning strategies by systematically analyzing the impact of key parameters and architectural components. Our experiments on the VR-Market dataset provided compelling evidence of our model's superior performance compared to state-of-the-art approaches. The best results are highlighted in bold.

6.1. *Impact of SR model & D-Noising(D-n) in VarReID*

The integration of an SR network and a D-n module significantly enhanced the performance of our PReID system. By improving image quality and feature extraction, these techniques led to substantial gains in accuracy and precision. On the VR-Market dataset, visualized in Fig. 6, the integration of the SR network increased Rank-1 accuracy by 3.35%, Rank-5 accuracy by 3.85%, and mAP by 3.22%. The D-n module further optimized the SR network, resulting in additional performance improvements of 2.19% in Rank-1 accuracy, 0.33% in Rank-5 accuracy, and 2.32% in mAP, as detailed in Table 6.

6.2. *Parameters variation in VarReID*

We demonstrated in our ablation research that integrating LW-up, RE, and CL into the ResNet-50 architecture resulted in notable performance improvements on the

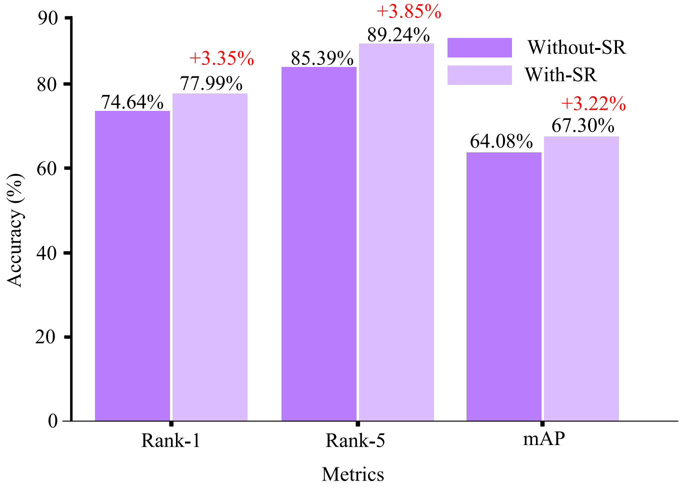


Fig. 6. Performance metrics without SR and with the SR network integration.

Table 6. Effect of the D-n module in the SR branch on VR-Market1501 (WON: without D-n; WN: with D-n).

Metric	WON	WN	Gain
Rank-1	75.80%	**77.99%**	+2.19%
Rank-5	85.48%	**85.81%**	+0.33%
Top-10	88.57%	**89.01%**	+0.44%
mAP	64.99%	**67.31%**	+2.32%

VR-Market dataset. By combining these techniques, in terms of Rank-1 (%), a significant improvement over the ResNet50 Baseline, ResNet50+CL, ResNet50+LW-up, and ResNet50+RE of 6.78%, 5.93%, 6.17%, and 4.26%, respectively, was achieved. In terms of mAP (%), our proposed ResNet50+CL+LW-up+RE outperformed the ResNet50 Baseline by 11.45%, ResNet50+CL by 9.75%, ResNet50+LW-up by 10.33%, and ResNet50+RE by 6.31%. Table 7 shows the results on the VR-Market dataset.

An ablation study was conducted on the VarReID model to analyze the impact of different techniques on LR PReID. The results demonstrated the importance of D-n,

Table 7. Impact of parameter variations and results of the ResNet50 model on the VR-Market (RE; LW-up; CL).

Config	RE	LW-up	CL	Batch	LR	Rank-1	mAP
ResNet50 Baseline	×	×	×	8	0.02	72.68%	59.40%
ResNet50+CL	×	×	✓	8	0.02	73.28%	60.54%
ResNet50+LW-up	×	✓	×	8	0.02	73.37%	60.15%
ResNet50+RE	✓	×	×	8	0.02	74.58%	62.85%
ResNet50+CL+LW-up+RE	✓	✓	✓	8	0.02	**77.90%**	**67.08%**

SR, and optimized learning strategies. The ResNet-50 baseline model was employed to further evaluate the effectiveness of RE, LW-up, and CL in improving performance.

6.3. *Case study*

This subsection presents representative case studies to provide deeper insight into VarReID beyond aggregate benchmark scores. Specifically, we examine (i) training dynamics across epochs and (ii) qualitative retrieval robustness under LR conditions, together with the complementary analyses in the evaluation of LR images and the component-wise resolution sensitivity under progressive degradation.

Case 1: Training Dynamics and Qualitative Retrieval Robustness.

Training dynamics across epochs: We analyze VarReID on VR-Market1501 using ResNet-50 as a baseline. As reported in Table 8, VarReID consistently outperforms ResNet-50 across epochs in Rank-1, Rank-5, and mAP. At epoch 90, VarReID improves mAP by 24.42 percentage points and Rank-1 by 19.03 percentage points relative to ResNet-50. Figure 7 further illustrates the performance trajectory over

Table 8. Epoch-wise comparison of ResNet50 and VarReID on VR-Market1501.

	ResNet50			VarReID		
Epoch	Rank-1	Rank-5	mAP	Rank-1	Rank-5	mAP
50	62.62%	76.07%	46.89%	**67.22%**	**80.05%**	**53.40%**
60	59.06%	74.44%	42.47%	66.06%	**80.29%**	52.55%
70	65.11%	77.49%	48.82%	**73.75%**	84.50%	62.49%
80	63.98%	76.66%	47.88%	**77.90%**	**89.39%**	**67.08%**
90	56.03%	71.35%	40.00%	75.06%	84.86%	64.42%
100	64.61%	77.85%	48.50%	75.92%	85.57%	65.21%

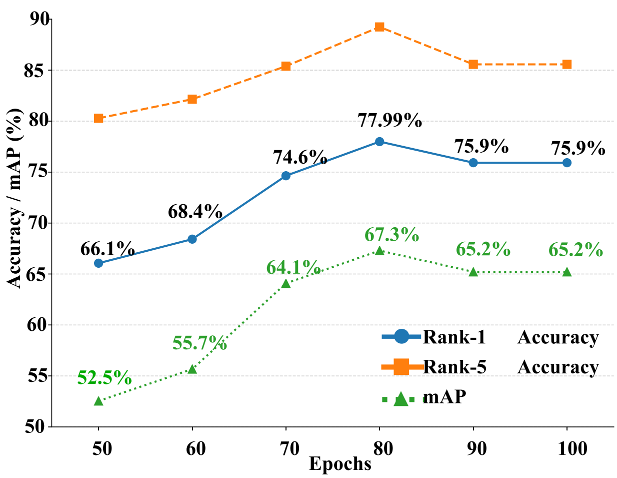


Fig. 7. Performance metrics across epochs (50–100) on the VR-Market1501 dataset.

epochs 50–100, highlighting sustained gains during training and peak performance at epoch 80 (Rank-1: 77.99%, mAP: 67.30%).

Qualitative retrieval robustness: Figs. 8 and 9 provide qualitative retrieval examples on VR-Market1501 and VR-MSMT17, respectively, showing each query with its top-10 retrieved candidates. Correct matches are highlighted in green and incorrect matches in red. Despite substantial resolution degradation and appearance variation, VarReID retrieves identity-consistent samples with improved robustness, indicating that the proposed pipeline enhances discriminative feature extraction under LR conditions. These qualitative results complement the quantitative benchmarks and support the practical applicability of VarReID in surveillance-style settings where LR imagery is common.

Case 2: Evaluation of Low-Resolution Images.

To assess resolution sensitivity, experiments were conducted on the Market-1501 dataset, where the resolution was progressively reduced to 32×64 (half), 16×32 (quarter), and 8×16 (eighth). As shown in Fig. 10, the model's performance

Fig. 8. VR-Market1501 sample retrieval results using VarReID.

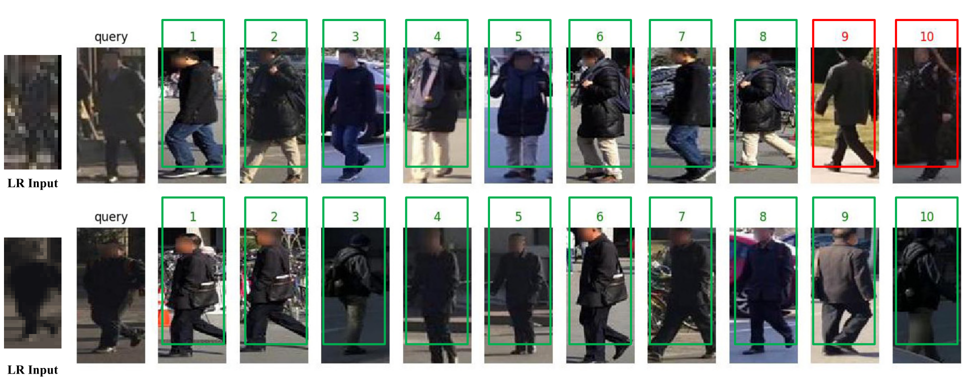


Fig. 9. VR-MSMT17 sample retrieval results using VarReID.

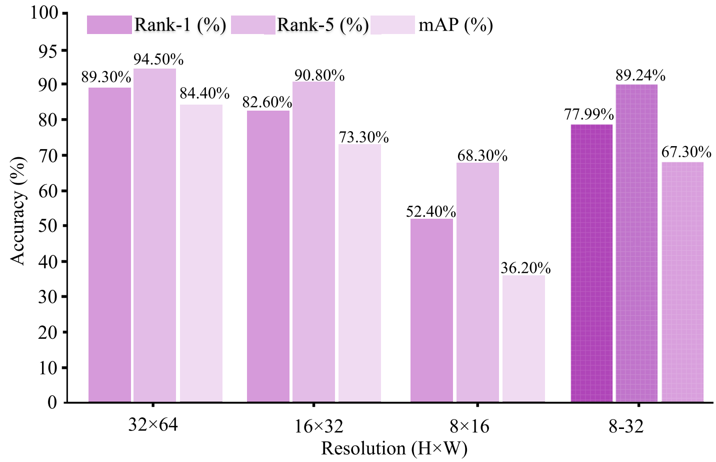


Fig. 10. Performance comparison of VarReID across resolution variations on the VR-Market dataset.

declined as the resolution decreased. Specifically, at 32×64, the Rank-1 accuracy reached 89.34%, but dropped to 82.63% at 16×32 and 52.43% at 8×16. Similarly, the Rank-5 accuracy fell from 94.54% at 32×64 to 90.83% at 16×32 and 68.29% at 8×16. The mAP score followed the same trend, decreasing from 84.42% at 32×64 to 73.35% at 16×32 and 36.22% at 8×16.

This sharp decline motivates a deeper examination of the underlying causes. It is important to note that the low-resolution samples were synthetically generated from the dataset's standard high-resolution images by downscaling them to 1/2, 1/4, and 1/8 of their original size. The severe performance drop at extreme resolutions (e.g. 8×16) is primarily due to the loss of critical identity cues. At such tiny spatial scales, essential visual attributes such as clothing texture, body contours, color gradients, and edge structure are heavily compressed or entirely lost, making it difficult for both the SR module and the PReID backbone to extract meaningful discriminative features. Future work may address these limitations by integrating stronger semantic priors, for example, pose guidance; leveraging multi-frame reconstruction to recover identity cues unavailable in single images; and utilizing larger, resolution-diverse datasets to better train models for these extremely low-resolution scenarios.

In contrast, the VR-Market dataset contains images with varying resolutions in the 8–32 range, where the model achieved a Rank-1 accuracy of 77.99%, a Rank-5 accuracy of 89.24%, and an mAP of 67.30%. This indicated that the model retained a notable level of discriminative ability even at lower resolutions.

These results underline the critical role of resolution in model performance and the trade-off between computational efficiency and recognition accuracy. The ability of the model to sustain performance across the 8–32 resolution range in VR-Market demonstrates its robustness in handling natural variations in image resolution.

Table 9. Resolution-sensitivity analysis quantifying the incremental effects of SR and denoising on PReID performance from moderate to extreme low-resolution inputs.

Resolution	Method	Rank-1	Rank-5	mAP	↑R1	↑R5	↑mAP
32×64	ResNet-50	79.44	89.91	63.37	—	—	—
	ResNet-50+SR	87.05	93.10	79.62	+7.61	+3.19	+16.25
	ResNet-50+SR+D-n	**89.30**	**94.50**	**84.40**	**+9.86**	**+4.59**	**+21.03**
16×32	ResNet-50	69.92	82.10	54.45	—	—	—
	ResNet-50+SR	78.74	89.57	67.55	+8.82	+7.47	+13.10
	ResNet-50+SR+D-n	**82.60**	**90.80**	**73.30**	**+12.68**	**+8.70**	**+18.85**
8×16	ResNet-50	42.31	61.69	28.47	—	—	—
	ResNet-50+SR	49.92	67.73	34.67	+7.61	+6.04	+6.20
	ResNet-50+SR+D-n	**52.40**	**68.30**	**36.20**	**+10.09**	**+6.61**	**+7.73**

Case 3: Resolution Sensitivity and Component-wise Impact Analysis.

To quantify resolution sensitivity and disentangle the contribution of each module under progressively degraded inputs, we downsample Market1501 to three resolution settings (32×64, 16×32, and 8×16). We evaluate a ResNet-50 baseline PReID model and two augmented variants: ResNet-50+SR and ResNet-50+SR+D-n (full VarReID). Table 9 reports the results, where ↑ denotes percentage-point gains over the baseline within the same resolution.

As summarized in Table 9, consistent trends emerge from moderate to extreme LR regimes. At 32×64 and 16×32, introducing SR produces the largest gains, particularly in mAP, indicating that SR recovers coarse, identity-relevant structure suppressed by downsampling. Adding D-n on top of SR yields further improvements across all metrics, increasing mAP to 84.40 at 32×64 and 73.30 at 16×32. This suggests that denoising attenuates SR-induced artifacts and stabilizes downstream feature extraction, resulting in more reliable retrieval.

In the extreme 8×16 corner case, all methods degrade markedly due to severe information loss; nevertheless, VarReID remains consistently superior to both the baseline and ResNet-50+SR, achieving gains of +10.09 Rank-1 and +7.73 mAP over the baseline. Although SR continues to provide measurable benefits, the reduced absolute performance underscores an inherent limitation: when fine-grained identity cues (e.g. texture, contour detail, and subtle color patterns) are largely absent, reconstruction alone cannot fully recover identity-discriminative evidence. These findings clarify the SR–PReID trade-off in VarReID: SR primarily restores coarse visual structure, whereas D-n improves feature reliability. The results also provide quantitative support for our cross-resolution design rationale, namely that the RAU enforces a canonical input scale prior to SR and PReID, thereby mitigating feature misalignment across heterogeneous resolutions.

7. Conclusion

In this work, we presented VarReID, an adaptive framework for low-resolution person re-identification (LR-PReID) in realistic surveillance settings. Although

VarReID builds on established components, its novelty lies in their adaptive integration and systematic evaluation across resolution regimes.

Experiments on MLR-DukeMTMC-reID, VR-Market1501, and VR-MSMT17 show that VarReID consistently outperforms representative LR-PReID methods. On VR-Market1501, adding SR improves Rank-1, Rank-5, and mAP by 3.35%, 3.85%, and 3.22%, respectively. The resolution sensitivity study indicates strong performance at 32×64 (89.34% Rank-1), but a marked drop at 8×16 (52.43%), highlighting the challenge of recovering identity cues under severe compression. The SR module in VarReID is optimized for visual reconstruction rather than identity discrimination, which may limit gains under extreme resolutions (e.g. 8×16) where discriminative cues are severely degraded. Future work will explore identity-aware SR and joint SR–PReID optimization to better preserve identity-relevant details under severe degradation.

Beyond surveillance, VarReID is applicable to smart retail, autonomous robotics, and low-bandwidth IoT scenarios where recognition from degraded imagery is required. Future research will enhance the SR pipeline via semantic-aware reconstruction, transformer-based contextual modeling, and multi-frame enhancement to improve robustness in ultra-low-resolution settings.

Declarations

- **Conflict of interest.** The authors declare that they have no known competing financial interests or personal relationships appeared to influence the work reported.
- **Competing interests.** The authors declare no competing interests.
- **Author Contributions:** TAA was responsible for the conceptualization, methodology design, data analysis, and paper writing. PZ and QM contributed to the conceptualization and provided supervision throughout the research. SM contributed to the review and editing of the paper.

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ORCID

Tariq Ali Arain https://orcid.org/0009-0006-5681-4830
Pengcheng Zhang https://orcid.org/0000-0003-3594-408X
Sehrish Mazhar https://orcid.org/0009-0000-7008-3102
Qing Meng https://orcid.org/0000-0001-7180-5736

References

1. M. Adil, S. Mamoon, A. Zakir, M. A. Manzoor and Z. Lian, Multi scale-adaptive super-resolution person re-identification using GAN, *IEEE Access* **8** (2020) 177351–177362.
2. T. A. Arain, P. Zhang, Q. Meng and A. U. Muhammad, Swin transformer with attention mechanism: A novel framework for person re-identification, *Pattern Anal. Appl.* **28**(2) (2025) 95.
3. E. B. Baoues, I. Jegham, S. Ameur and A. B. Khalifa, GF2PREID: A novel framework for person re-identification using generative networks, in *2023 Int. Conf. Cyberworlds (CW)* (IEEE, 2023), pp. 102–109.
4. X. Chang, T. M. Hospedales and T. Xiang, Multi-level factorisation net for person re-identification, in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)* (IEEE, 2018), pp. 2109–2118.
5. S. W. Park, S. H. Jung and B. Chun, Correction: NextSRGAN: Enhancing super-resolution GAN with ConvNeXt discriminator for superior realism, *Vis Comput.* **41** (2025) 8119.
6. J. Crawford, H. Yin, L. McDermott and D. Cummings, UniCat: Crafting a stronger fusion baseline for multimodal re-identification, preprint (2023), arXiv:2310.18812.
7. L. Du, J. Qi, T. Gao, Y. Gao and D. Liu, A cooperative network for low-resolution person re-identification, in *2024 4th Int. Conf. Comput. Commun. Artif. Intell. (CCAI)* (IEEE, 2024), pp. 70–74.
8. Y. Ge *et al.*, FD-GAN: Pose-guided feature distilling GAN for robust person re-identification, *Advances in Neural Information Processing Systems*, Vol. 31 (NeurIPS, 2018).
9. P. Ghadekar, A. Kabra, K. Gangwal, A. Kinage, K. Agarwal and K. Chaudhari, Improving person-re identification with dual-generator and dual-discriminator architecture in conditional GANs, in *2024 1st Int. Conf. Cogn. Green Ubiquitous Comput. (IC-CGU)* (IEEE, 2024), pp. 1–9.
10. K. Han, Y. Huang, Z. Chen, L. Wang and T. Tan, Prediction and recovery for adaptive low-resolution person re-identification, in *Computer Vision-ECCV 2020*, eds. A. Vedaldi, H. Bischof, T. Brox and J. M. Frahm, Lecture Notes in Computer Science, Vol. 12371, (Springer, Cham. 2020) pp. 193–209.
11. K. He, X. Zhang, S. Ren and J. Sun, Deep residual learning for image recognition, in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)* (IEEE, 2016), pp. 770–778.
12. J. Hu, L. Shen and G. Sun, Squeeze-and-excitation networks, in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)* (IEEE, 2018), pp. 7132–7141.
13. G. Huang, Z. Liu, L. Van Der Maaten and K. Q. Weinberger, Densely connected convolutional networks, in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)* (IEEE, 2017), pp. 4700–4708.
14. M. Huang, C. Hou, X. Zheng and Z. Wang, Multi-resolution feature perception network for UAV person re-identification, *Multimed. Tools Appl.* **83** (2024) 62559–62580.
15. J. Jiao, W.-S. Zheng, A. Wu, X. Zhu and S. Gong, Deep low-resolution person re-identification, in *Proc. AAAI Conf. Artif. Intell.* 32(1) (2018), pp. 6967–6974.
16. X. Jin, C. Lan, W. Zeng, G. Wei and Z. Chen, Semantics-aligned representation learning for person re-identification, in *Proc. AAAI Conf. Artif. Intell.* **34**(7) (2020) 11173–11180.
17. C. Ledig *et al.*, Photo-realistic single image super-resolution using a generative adversarial network, in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)* (IEEE, 2017), pp. 4681–4690.
18. W. Li, X. Zhu and S. Gong, Harmonious attention network for person re-identification, in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)* (IEEE, 2018), pp. 2285–2294.

19. Y.-J. Li, Y.-C. Chen, Y.-Y. Lin, X. Du and Y.-C. F. Wang, Recover and identify: A generative dual model for cross-resolution person re-identification, in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)* (IEEE, 2019), pp. 8090–8099.
20. Y.-J. Li, Y.-C. Chen, Y.-Y. Lin and Y.-C. F. Wang, Cross-resolution adversarial dual network for person re-identification and beyond, preprint (2020), arXiv:2002.09274.
21. T. Mamedov, A. Konushin and V. Konushin, Remix: Training generalized person re-identification on a mixture of data, in *2025 IEEE/CVF Winter Conf. Appl. Comput. Vis. (WACV)* (IEEE, 2025), pp. 8186–8196.
22. S. Mao, S. Zhang and M. Yang, Resolution-invariant person re-identification, preprint (2019), arXiv:1906.09748.
23. S. H. Park, Y. S. Moon and N. I. Cho, Perception-oriented single image super-resolution using optimal objective estimation, in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)* (IEEE, 2023), pp. 1725–1735.
24. S. S. Pereira and J. E. B. Maia, MC-MIL: Video surveillance anomaly detection with multi-instance learning and multiple overlapped cameras, *Neural Comput. Appl.* **36**(18) (2024) 10527–10543.
25. X. Qian, Y. Fu, T. Xiang, W. Wang, J. Qiu, Y. Wu, Y.-G. Jiang and X. Xue, Pose-normalized image generation for person re-identification, in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, eds. V. Ferrari, M. Hebert, C. Sminchisescu, Y. Weiss, Lecture Notes in Computer Science, Vol. 11213 (Springer, Cham. 2018), pp. 650–667.
26. O. J. Santana, J. Lorenzo-Navarro, D. Freire-Obregón, D. Hernández-Sosa and M. Castrillón-Santana, Applying deep learning image enhancement methods to improve person re-identification, *Neurocomputing* **598** (2024) 128011.
27. Y. Shen, T. Xiao, H. Li, S. Yi and X. Wang, End-to-end deep Kronecker-product matching for person re-identification, in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)* (IEEE, 2018), pp. 6886–6895.
28. V. Somers, C. De Vleeschouwer and A. Alahi, Body part-based representation learning for occluded person re-identification, in *Proc. IEEE/CVF Winter Conf. Appl. Comput. Vis. (WACV)* (IEEE, 2023), pp. 1613–1623.
29. Y. Suh, J. Wang, S. Tang, T. Mei and K. M. Lee, Part-aligned bilinear representations for person re-identification, in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, eds. V. Ferrari, M. Hebert, C. Sminchisescu, Y. Weiss, Lecture Notes in Computer Science, Vol. 11218 (Springer, Cham. 2018), pp. 402–419.
30. L. Sun, J. Liang, S. Liu, H. Yong and L. Zhang, Perception-distortion balanced super-resolution: A multi-objective optimization perspective, *IEEE Trans. Image Process.* **33** (2024) 4444–4458.
31. Y. Sun, L. Zheng, W. Deng and S. Wang, SVDNet for pedestrian retrieval, in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)* (IEEE, 2017), pp. 3800–3808.
32. Y. Sun, L. Zheng, Y. Yang, Q. Tian, S. Wang, J. Xiong and X. Li, Beyond part models: Person retrieval with refined part pooling (and a strong convolutional baseline), in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, eds. V. Ferrari, M. Hebert, C. Sminchisescu, Y. Weiss, Lecture Notes in Computer Science, Vol. 11208 (Springer, Cham. 2018), pp. 266–283.
33. M. Wang, Z. Li, H. Liu, Z. Chen and K. Cai, SP-IGAN: An improved GAN framework for effective utilization of semantic priors in real-world image super-resolution, *Entropy* **27**(4) (2025) 414.
34. Q. Wang, K. Jiang, Z. Wang, W. Ren, J. Zhang and C.-W. Lin, Multi-scale fusion and decomposition network for single image deraining, *IEEE Trans. Image Process.* **33** (2023) 191–204.
35. Z. Wang, M. Ye, F. Yang, X. Bai and S. Satoh, Cascaded SR-GAN for scale-adaptive low resolution person re-identification, in *Int. Joint Conf. Artificial Intelligence.* **1** (2018) 4.

36. D. Wu, Z. Liu, Z. Chen, S. Gan, K. Tan, Q. Wan and Y. Wang, LRMM: Low rank multi-scale multi-modal fusion for person re-identification based on RGB-NI-TI, *Expert Syst. Appl.* **263** (2025) 125716.
37. L. Y. Wu, L. Liu, Y. Wang, Z. Zhang, F. Boussaid, M. Bennamoun and X. Xie, Learning resolution-adaptive representations for cross-resolution person re-identification, *IEEE Trans. Image Process.* **32** (2023) 4800–4811.
38. J. Xie, Y. Ge, J. Zhang, S. Huang, F. Chen and H. Wang, Low-resolution assisted three-stream network for person re-identification, *Vis. Comput.* **38** (2022) 2515–2525.
39. X. Xu, X. Yuan, Z. Wang, K. Zhang and R. Hu, Rank-in-rank loss for person re-identification, *ACM Trans. Multimedia Comput. Commun. Appl.* **18**(2s) (2022) 1–21.
40. Z. Yang, M. Lin, X. Zhong, Y. Wu and Z. Wang, Good is bad: Causality inspired cloth-debiasing for cloth-changing person re-identification, in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)* (IEEE, 2023), pp. 1472–1481.
41. C. Yuan, Z. Guan, Y. Xu, X. Chen, X. Zhu and W. Liang, Jointly adaptive cross-resolution person re-identification on super-resolution, *Complex Intell. Syst.* **11**(6) (2025) 249.
42. X. Yuan, X. Xu, Z. Wang, K. Zhang, W. Liu and R. Hu, Searching parameterized retrieval & verification loss for re-identification, *IEEE J. Sel. Top. Signal Process.* **17**(3) (2023) 560–574.
43. L. Zhang, Y. Xu, L. Zhao and F. Qin, Resolution independent person re-identification network, *IET Comput. Vis.* **19**(1) (2025) e12140.
44. Y. Zhang, S. Lee and A. Yao, Pairwise distance distillation for unsupervised real-world image super-resolution, in *Eur. Conf. Comput. Vis. (ECCV)*, eds. A. Leonardis, E. Ricci, S. Roth, O. Russakovsky, T. Sattler, G. Varol, Lecture Notes in Computer Science, Vol. 15087 (Springer, Cham. 2024), pp. 429–446.
45. Z. Zhang, L. Wan, W. Xu and S. Wang, Low-resolution human pose estimation and action recognition via pose-driven super-resolution reconstruction, *Mach. Learn.* **114**(6) (2025) 135.
46. A. Zheng, Z. Wang, Z. Chen, C. Li and J. Tang, Robust multi-modality person re-identification, in *Proc. AAAI Conf. Artif. Intell.* **35**(4) (2021) 3529–3537.
47. W.-S. Zheng, J. Hong, J. Jiao, A. Wu, X. Zhu, S. Gong, J. Qin and J. Lai, Joint bilateral-resolution identity modeling for cross-resolution person re-identification, *Int. J. Comput. Vis.* **130** (2022) 136–156.
48. W.-S. Zheng, J. Yan and Y.-X. Peng, A versatile framework for multi-scene person re-identification, *IEEE Trans. Pattern Anal. Mach. Intell.* **47**(3) (2025)1362–1380.
49. Z. Zheng, L. Zheng and Y. Yang, Pedestrian alignment network for large-scale person re-identification, *IEEE Trans. Circuits Syst. Video Technol.* **29**(10) (2018) 3037–3045.
50. Z. Zhong, L. Zheng, G. Kang, S. Li and Y. Yang, Random erasing data augmentation, in *Proc. AAAI Conf. Artif. Intell.* **34** (2020) 13001–13008.
51. X. Zhou, J. Liu, X. Jiang, P. Li and C. Zhao, Multi-clues adaptive learning for cloth-changing person re-identification, *Int. J. Pattern Recognit. Artif. Intell.* **39**(6) (2025) 2555001.
52. F. Zhu, D. Li, C. Sun and B. Zhu, Cross-resolution person re-identification based on double-layer graph convolution network, *Eng. Appl. Artif. Intell.* **147** (2025) 110376.

Tariq Ali Arain received the Master of Engineering degree in Computer Applied Technology from Nanjing University of Science and Technology, China, and the Ph.D. degree in Software Engineering from Hohai University, Nanjing, China. His research interests include computer vision, person re-identification, deep learning, low-resolution recognition, image processing, transformer-based architectures, and robust modeling for complex real-world environments. His current work focuses on advancing multimodal and text-based retrieval for person re-identification by integrating vision–language learning and lightweight knowledge-enhanced reasoning to improve robustness, efficiency, and interpretability. He has authored and co-authored several papers in SCI-indexed journals and CCF-ranked conferences, and has also served as a peer reviewer for SCI-indexed journals.

Pengcheng Zhang received the Ph.D. degree in computer science from Southeast University, China. He is currently a Professor and Vice Dean with the College of Computer Science and Software Engineering, Hohai University, Nanjing, China, where he serves as a Ph.D. and master's supervisor. His research interests include software engineering, service computing, data science, hydrological big data mining, and computer vision. He has led and participated in numerous research projects funded by national and provincial agencies, including the National Natural Science Foundation of China. He has published extensively in international journals, including IEEE Transactions on Services Computing, IEEE Transactions on Big Data, and Information and Software Technology. His recent research also includes person re-identification for intelligent surveillance systems.

Sehrish Mazhar received the Master of Science degree in Mathematics from Hohai University, Nanjing, China, and M.Sc. and B. Sc. degrees from Government College University, Faisalabad, Pakistan. She is currently focusing on research in computer vision and autonomous systems. Her research interests include person re-identification (P-ReID), deep learning, object detection, scene understanding, and intelligent navigation for autonomous systems. She has authored and coauthored several research articles in international SCI-indexed journals and conferences.

Qing Meng received the B.S. degree from East China Normal University, China, in 2012, and the M.S. and Ph.D. degrees from Southeast University, China, in 2017 and 2022, respectively. He is currently a Lecturer with the College of Computer Science and Software Engineering, Hohai University, Nanjing, China. His research interests include social media analytics, recommender systems, graph neural networks, and computer vision. He is the principal investigator of an NSFC Youth Science Fund Project and has published in leading international journals and conferences. His recent research also includes person re-identification for intelligent surveillance systems.